Peer Pressure in Linear-Quadratic Games

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Motivation

Social interactions shape outcomes in schools, workplaces and social networks.

(Passive) peer effects

- The return to my action depends on what my peers do.
- When my friends study, my own studying pays off more.

But people also actively try to change what their peers do

- Students encourage friends to study (or to skip class).
- Coworkers pressure colleagues to meet a given goal.

Terminology

Peer effects = passive, complementarity in actions.
Peer pressure = active, a costly action taken to change others' incentives.

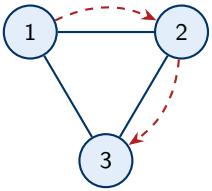
This paper

Standard LQ network games

- Influence is passive (Ballester, Calvó-Armengol & Zenou, 2006; Bramoullé, Kranton & D'Amours, 2014).

We add costly, strategic peer pressure

- Friendship network **G** governs peer effects in effort.
- Peer-pressure network **H** says who can pressure whom.
- Agents choose effort *and* how hard to pressure each peer.



friendship **G** (solid)

pressure **H** (dashed)

Question

How does the possibility of exerting peer pressure change equilibrium behavior, welfare, and what we can learn from data?

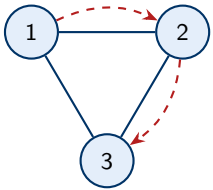
Related literature

Plan of the rest of the talk

1. **Model.** Effort and peer pressure on two networks, friendship **G** and peer pressure **H**, in a two-stage game.
2. **Equilibrium.** Closed-form characterization and its intuition, an example where the effort ranking flips, how pressure changes interdependence, and complementarity of pressure decisions.
3. **Welfare.** The planner's problem, underprovision, the link to targeting (Galeotti, Golub & Goyal, 2020). (time permitting).
4. **Empirical implications** (work in progress)**.** What effort data reveal, and how to separate pressure from passive peer effects.
5. **Extensions and conclusion.**

Model: players and networks

- Set N of n players.
- Each player i chooses
  - effort $x_i \geq 0$ (e.g. studying),
  - pressure $b_{ij} \geq 0$ on each peer j (e.g. encouragement to study).



friendship **G** (solid)

pressure **H** (dashed)

Two networks (binary, no self-loops, possibly directed)

- Friendship $\mathbf{G} = (g_{ij})$. $g_{ij} = 1$ if i's return to effort depends on x_j.
- Peer pressure $\mathbf{H} = (h_{ij})$. $h_{ij} = 1$ if i can pressure j.
- **H** need not coincide with **G**; $\mathbf{H} = \mathbf{G}$ is a special case.

Notation. $\mathbf{B} = (b_{ij})$, and $z_j = \sum_i h_{ij} b_{ij}$ is the pressure j receives.

Model: payoffs

Utility

$$u_i(\boldsymbol{x}, \mathbf{B}) = x_i \Big[a_i \; + \; \delta \sum_j h_{ji} b_{ji} \; + \; \alpha \sum_j g_{ij} x_j \Big] \; - \; \tfrac{1}{2} x_i^2 \; - \; \tfrac{1}{2} \sum_j b_{ij}^2.$$

Marginal return to effort x_i has three components:

- a_i: stand-alone return.
- $\alpha \sum_j g_{ij} x_j$: standard peer effects via $\mathbf{G}$. Assume $\alpha > 0$: strategic complements.
- $\delta \sum_j h_{ji} b_{ji}$: new channel: pressure received via $\mathbf{H}$ ($\delta > 0$).

Important feature: b_{ij} is *privately costly* but yields *no direct benefit*. Its value arises only through the induced increase in x_j, which spreads through $\mathbf{G}$ and feeds back into i's payoff.

Model: timing and equilibrium concept

Two-stage game

Stage 1: Players simultaneously choose peer pressure $\mathbf{B} = (b_{ij})_{i,j \in N}$.
Stage 2: Given $\mathbf{B}$, players simultaneously choose effort $\boldsymbol{x} = (x_i)_{i \in N}$.

Solution concept: subgame-perfect Nash equilibrium.

Two technical assumptions:

A1. $\rho(\alpha \mathbf{G}) < 1$: the standard LQ condition; guarantees that $\mathbf{M} \equiv (\mathbf{I} - \alpha \mathbf{G})^{-1} = (m_{ij})$ exists.

A2. $\rho\big(\delta^2 \mathbf{M}(\mathbf{M} \circ \mathbf{H})'\big) < 1$: ensures strict concavity of stage-1 payoffs in $(b_{ij})_{j \in N}$ and a unique stage-1 solution.

($\circ$ = Hadamard product; $\mathbf{M} \circ \mathbf{H}$ keeps m_{ij} on pressure links, $h_{ij} = 1$, and is zero elsewhere.)

Equilibrium characterization

Proposition (equilibrium)

Under A1 and A2, a unique SPE exists. Equilibrium effort and pressure are:

$$\boldsymbol{x}^* = \boldsymbol{x}^*(\delta) = \left[\mathbf{I} - \delta^2\, \mathbf{M}(\mathbf{M} \circ \mathbf{H})'\right]^{-1} \mathbf{M}\, \boldsymbol{a},$$

$$\mathbf{B}^* = \delta\ \mathrm{diag}(\boldsymbol{x}^*)\,(\mathbf{M} \circ \mathbf{H}), \qquad \text{i.e.} \qquad b_{ij}^* = \delta\, h_{ij}\, m_{ij}\, x_i^*.$$

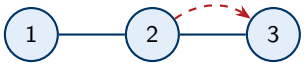
When $\delta = 0$: $\boldsymbol{x}^*(0) = \mathbf{M}\boldsymbol{a}$, recovering the standard LQ model.

When $\delta > 0$:

$$\boldsymbol{x}^*(\delta) - \boldsymbol{x}^*(0) = \overbrace{\left[\sum_{t=1}^{\infty} \delta^{2t} \left(\mathbf{M}\,(\mathbf{M} \circ \mathbf{H})'\right)^t\right]}^{\text{Additional network effect}} \overbrace{\mathbf{M}\boldsymbol{a}}^{=\boldsymbol{x}^*(0)}.$$

Peer pressure creates additional network multipliers in addition to the standard Katz–Bonacich centrality.

Example: the effort ranking can flip



friendship **G** (solid)

pressure **H** (dashed)

	x_1^*	x_2^*	x_3^*	b_{23}^*
$\delta = 0$	1.59	**1.95**	1.59	0
$\delta = 1$	1.68	2.25	**2.50**	0.82

$\boldsymbol{a} = \mathbf{1}$, $\alpha = 0.3$. A2 requires $\delta < 2.73$.

- Without pressure, the center (player 2) exerts the most effort.
- With pressure along $2 \to 3$, player 3 overtakes player 2 for every $\delta \in (0.79, 2.73)$. Player 2 exerts all the pressure, player 3 the most effort.
- Even with $\mathbf{H} = \mathbf{G}$ undirected, pressure can overturn the Katz–Bonacich ranking (example in the paper).

Some intuition: $\boldsymbol{x}^*(\delta) - \boldsymbol{x}^*(0)$

With $\boldsymbol{x}^*(0) = \mathbf{M}\boldsymbol{a}$:

$$\boldsymbol{x}^*(\delta) - \boldsymbol{x}^*(0) = \delta\,\mathbf{M}\boldsymbol{z}^* \qquad \text{and} \qquad \boldsymbol{z}^* = \delta\,(\mathbf{M} \circ \mathbf{H})'\,\boldsymbol{x}^*(\delta).$$

- **Effort from pressure.** The effort due to pressure is the pressure received, $z_i^* = \sum_j h_{ji} b_{ji}^*$, propagated through the friendship network.
- **Pressure from effort.** j pressures i in proportion to her own effort, $b_{ji}^* = \delta\, h_{ji} m_{ji} x_j^*$.

Other way to look at it:

$$\boldsymbol{x}^*(\delta) - \boldsymbol{x}^*(0) = \sum_{t=1}^{\infty} \delta^{2t} \left(\mathbf{M}(\mathbf{M} \circ \mathbf{H})'\right)^t \boldsymbol{x}^*(0).$$

- **One round,** $\delta^2[\mathbf{M}(\mathbf{M} \circ \mathbf{H})']_{kj} = \sum_\ell \delta\, m_{k\ell} \cdot \delta\, h_{j\ell} m_{j\ell}$. A unit of j's effort makes j pressure each ℓ she can pressure by $\delta h_{j\ell} m_{j\ell}$, which raises k's effort by $\delta m_{k\ell}$.
- **Round** t**.** From $\boldsymbol{x}^*(0)$, effort triggers pressure, pressure raises effort, and so on. α (through $\mathbf{M}$) and δ^2 act as discount factors.

How pressure changes interdependence

Interdependence $L_{ij}(\delta) = \partial x_i^* / \partial a_j$. Without pressure $\mathbf{L}(0) = \mathbf{M}$.

Proposition (local approximation, $\mathbf{H} = \mathbf{G}$, possibly directed)

$$\mathbf{L}(\delta) - \mathbf{L}(0) = \alpha\delta^2\,\mathbf{G}' + \alpha^2\delta^2\big[\mathbf{G}\mathbf{G}' + \mathbf{G}'\mathbf{G} + (\mathbf{G}^2 \circ \mathbf{G})'\big] + \alpha^2\delta^4(\mathbf{G}')^2 + O(\alpha^3).$$

→ $g_{ab} = 1$, a's return depends on x_b ⇢ $h_{ab} = 1$, a can pressure b (with $\mathbf{H} = \mathbf{G}$ each link is both; color = role)

$\alpha\delta^2\,\mathbf{G}'$		x_j rises, so j pressures i more. The *target* responds, hence $\mathbf{G}'$.
$\alpha^2\delta^2\,\mathbf{G}\mathbf{G}'$		j pressures k, whose extra effort raises i's ($i \to k$).
$\alpha^2\delta^2\,\mathbf{G}'\mathbf{G}$		k gains from j's effort ($k \to j$), then pressures i more.
$\alpha^2\delta^2\,(\mathbf{G}^2 \circ \mathbf{G})'$		he link $j \to i$ closed by the path $j \to k \to i$. Pushing i raises x_i, then x_k, then x_j, so j pressures i even harder.
$\alpha^2\delta^4\,(\mathbf{G}')^2$		Second round. j pressures k, who then pressures i more, hence δ^4.

Take-away: Local feedback loops amplify the effects of peer pressure.

Welfare: the planner's problem

- Beyond the standard effort externality, a new inefficiency arises, since senders weigh only their own benefit from pressure.
- The planner chooses $\mathbf{B}$, efforts stay noncooperative (e.g. mentoring schemes, team composition).

$$\max_{\mathbf{B}} \; W(\mathbf{B}) = \tfrac{1}{2}\Big(\textstyle\sum_i x_i(\mathbf{B})^2 - \sum_{i,j} b_{ij}^2\Big)$$

A3 $\delta^2 \lambda_{\max}(\mathbf{MDM}') < 1$, where $\mathbf{D} = \operatorname{diag}(\boldsymbol{d})$ and d_j is the number of agents who can pressure j.

Proposition (planner)

Under A1 and A3,

$$\boldsymbol{x}^{SP} = \left[\mathbf{I} - \delta^2\,\mathbf{MDM}'\right]^{-1}\mathbf{M}\boldsymbol{a}, \qquad b_{ij}^{SP} = \delta\, h_{ij}\,[\mathbf{M}'\boldsymbol{x}^{SP}]_j.$$

- Only the pressure each agent *receives* matters, so it is split equally among her potential senders.
- Target j is valued by her social return $\sum_k m_{kj} x_k$; more potential senders make her cheaper.

Pressure is underprovided on every link

$$\text{private marginal value of } b_{ij}\text{: } \delta\, m_{ij}\, x_i \quad \neq \quad \text{social marginal value of } b_{ij}\text{: } \delta \sum_k m_{kj}\, x_k$$

- At $\mathbf{B}^*$ the gap is $\delta h_{ij} \sum_{k \neq i} m_{kj} x_k^* > 0$. The sender ignores the target's own gain and the gains of third parties.

Proposition (underprovision)

Under A1 and A3, $\boldsymbol{x}^{SP} \geq \boldsymbol{x}^*$, and on every link with $h_{ij} = 1$,

$$b_{ij}^{SP} - b_{ij}^* \; > \; 0.$$

- Everyone who can be pressured receives strictly more pressure from the planner.
- Not only a matter of scale. Equilibrium weights pressure by the sender's feedback $m_{ij} x_i$, the planner by the target's social value, equally across senders.

The planner's problem is a targeting problem

Galeotti, Golub & Goyal (2020, Ecta) [GGG]. A planner shifts stand-alone returns $\boldsymbol{a} \to \boldsymbol{a} + \boldsymbol{y}$, then agents play the LQ game.

$$\max_{\boldsymbol{y}} \ \tfrac{1}{2}\left\|\mathbf{M}(\boldsymbol{a}+\boldsymbol{y})\right\|^2 \quad \text{s.t.} \quad \|\boldsymbol{y}\|^2 \le C.$$

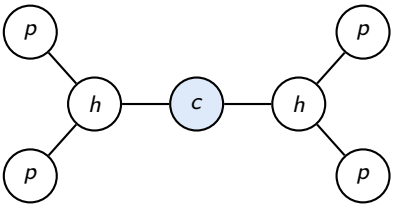
Proposition (equivalence)

Let $\mathbf{G}$ be symmetric and connected, $\mathbf{H} = \mathbf{J} - \mathbf{I}$, $\boldsymbol{a} \gg \mathbf{0}$, and A1, A3 hold. Exists $C = C_\delta$, such that the planner's pressure shifts stand-alone returns by exactly GGG's optimal intervention $\boldsymbol{y}^G$,

$$\delta\, \boldsymbol{z}(\mathbf{B}^{SP}) = \boldsymbol{y}^G, \qquad b_{ij}^{SP} = \frac{y_j^G}{\delta(n-1)}.$$

- Pressure acts on efforts only through δz_j, a shift in j's stand-alone return.
- Equal split among the $n-1$ senders is the cheapest delivery, at a cost proportional to GGG's. The shadow price $1/(2\delta^2(n-1))$ depends only on δ and n.
- Equilibrium pressure is a decentralized GGG intervention, chosen for private feedback.

Example: who is pressured the most?



Friendship **G**. Broker c, hubs h, leaves p.
$\mathbf{H} = \mathbf{J} - \mathbf{I}$, $\boldsymbol{a} = \mathbf{1}$, $\alpha = 0.3$, $\delta = \sqrt{6}/20$.

Pressure received z_j	broker c	hub h	leaf p
Equilibrium	**0.493**	0.460	0.267
Planner	12.79	**13.05**	6.80

- **Equilibrium.** The broker receives the most. Hubs exert the most effort, and pushing the broker raises a hub's own effort more than pushing her own leaf ($m_{hc} = 0.469 > m_{hp} = 0.417$).
- **Planner.** The hubs receive the most. The planner also counts the target's own gain $m_{jj}x_j$, which is larger for a hub (9.71 vs 8.66).
- With δ at $\frac{3}{4}$ of its A3 bound, the rankings differ for every $\alpha \in (0.202, 0.5)$; the planner always ranks the hubs first.

Empirical implications: challenges

work in progress

Eliminating the equilibrium pressure from the stage-2 first-order condition gives

$$x_i^* = a_i + \alpha \sum_j g_{ij} x_j^* + \delta^2 \sum_j h_{ji} m_{ji} x_j^*. \quad (1)$$

- i responds to her friends' efforts *and* to the efforts of those who can pressure her.
- m_{ij} is nonlinear function of α.
- (1) is the equilibrium of an LQ game *without* pressure, with interaction matrix $\alpha\mathbf{G} + \delta^2(\mathbf{M} \circ \mathbf{H})'$. Welfare implications are different.

Data

- Can we observe separtelly $\mathbf{H}$ and $\mathbf{G}$?
- If not, is $\mathbf{G} = \mathbf{H}$ the most plausible assumption?

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Empirical implications: approximating **M**

work in progress

Exact weights need the whole network and are nonlinear in α. Expanding $\mathbf{M} = \sum_s \alpha^s \mathbf{G}^s$,

$$x_i \approx a_i + \alpha \textstyle\sum_j g_{ij} x_j + \alpha\delta^2 \sum_j h_{ji} x_j, \tag{1A}$$

$$x_i \approx a_i + \alpha \textstyle\sum_j g_{ij} x_j + \alpha\delta^2 \sum_j h_{ji} x_j + \alpha^2\delta^2 \sum_j h_{ji} [\mathbf{G}^2]_{ji}\, x_j, \tag{1B}$$

and, when $\mathbf{H} = \mathbf{G}$ and $\mathbf{G}$ is symmetric,

$$x_i \approx a_i + \alpha(1+\delta^2) \textstyle\sum_j g_{ij} x_j + \alpha^2\delta^2 \sum_j g_{ij} [\mathbf{G}^2]_{ij}\, x_j. \tag{1C}$$

- **G and H observed** The pressure term is a separate regressor, so (1A) identifies α and $\alpha\delta^2$, hence δ. (1B) adds an additional testable restriction.
- $\mathbf{G} = \mathbf{H}$ (both symmetric. Common-friends term of (1C), is zero under the standard LQ model.
- **Note** A common-friends premium also fits passive complementarities that are stronger among friends who share friends (Jackson et al. (2012, AER); Karlan et al. (2009, QJE).

Extensions and conclusion

Extensions (both with a unique closed-form equilibrium)

- *Interdependent pressure decisions*, $u_i + \gamma \sum_{j \neq k} h_{ij} h_{ik} b_{ij} b_{ik}$. With $\gamma > 0$ (economies of scope) pressure is spread more evenly across targets; with $\gamma < 0$ (limited attention) it concentrates on those with the strongest feedback.
- *Pressure with a private return*, $u_i + \gamma x_i \sum_j h_{ij} b_{ij}$. Then $b^*_{ij} = x^*_i h_{ij} (\gamma m_{ii} + \delta m_{ij})$, so pressure is exerted even without complementarities ($\alpha = 0$).

Conclusion

- With costly, strategic pressure, effort is a centrality of the combined **G–H** structure, and pressure decisions are global complements.
- Pressure is underprovided on every link; the model is a non-cooperative version of Galeotti, Golub & Goyal, (2020, Ecta).
- Pressure and passive peer effects can generate the same efforts but differ in welfare. Testable restrictions?

Thank you!

Related literature

LQ network games (passive peer effects)

- Ballester, Calvó-Armengol & Zenou (Ecta, 2006); Calvó-Armengol, Patacchini & Zenou (REStud, 2009); Bramoullé, Kranton & D'Amours (AER, 2014).

Active peer pressure

- Calvó-Armengol & Jackson (JEEA, 2010). Binary actions, multiple equilibria, Pareto-perfect refinement.
- Shi (EER, 2024). Helping in teams, strategic substitutes, multiple equilibria.
- *Here.* Continuous intensities, a unique equilibrium, two distinct networks, and pressure decisions that are complements.

Targeting interventions

- Galeotti, Golub & Goyal (Ecta, 2020). A planner shifts stand-alone returns.
- *Here.* Agents shift each other's returns. When everyone can pressure everyone, our planner's problem maps exactly into theirs.

◂ Back to *This paper*

Pressure decisions are strategic complements

Reduced stage-1 payoff $U_i(\mathbf{B}) = \frac{1}{2}\big(x_i(\mathbf{B})^2 - \sum_j b_{ij}^2\big)$.

Proposition (supermodularity)

$$\underbrace{\frac{\partial^2 U_i}{\partial b_{ij}\,\partial b_{\ell k}} = \delta^2\, h_{ij} m_{ij}\, h_{\ell k} m_{ik} \;\geq 0,}_{\text{for all } (\ell,k)\neq(i,j)} \qquad \underbrace{\frac{\partial U_i}{\partial b_{\ell k}} = \delta\, h_{\ell k} m_{ik}\, x_i(\mathbf{B}) \;\geq 0\,.}_{\text{for all } \ell\neq i}$$

- Complementarity is global. Any pressure anywhere that raises x_i raises the return to all of i's pressure, which scales with x_i.
- A public good whose contributions are complements, not substitutes.

Monotone comparative statics

If α, δ, $\boldsymbol{a}$, $\mathbf{G}$ or $\mathbf{H}$ increase (with A1, A2 holding), then $\mathbf{B}^*$, $\boldsymbol{x}^*$ and *every* payoff u_i^* weakly increase.